

1 LDA vs. QDA Concept Questions

(a) LDA

From Bayes' theorem, we have

$$\Pr(Y = k | X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)}. \quad (1)$$

To find the class k corresponding to the maximum of the conditional probability, we only need to consider the numerator, since the denominator is the same for all k . Taking the logarithm of the numerator and assuming f_k is a Gaussian distribution with the same variance σ for every class, we have

$$\begin{aligned} \log(\pi_k f_k(x)) &= \log \pi_k + \log f_k(x) \\ &= \log \pi_k - \log(\sqrt{2\pi}\sigma) - \frac{(x - \mu_k)^2}{2\sigma^2}. \end{aligned} \quad (2)$$

Expand the squared term in (2), we have

$$\log \pi_k - \log(\sqrt{2\pi}\sigma) - \frac{x^2 - 2x\mu_k + \mu_k^2}{2\sigma^2} = \log \pi_k - \log(\sqrt{2\pi}\sigma) - \frac{x^2}{2\sigma^2} + \frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2}. \quad (3)$$

In Eq. 3, x is the observed value and does not depend on k . $\sqrt{2\pi}\sigma$ is a constant, since σ is shared by all classes. Therefore, the two terms $\log(\sqrt{2\pi}\sigma)$ and $x^2/(2\sigma^2)$ can be removed, as neither one affects which k maximizes the expression. Then, we have

$$\frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k. \quad (4)$$

Eq. 4 is the definition of discriminant function. It is easy to show that the class k that maximizes Eq. 4 is the same k that maximizes the conditional probability $\Pr(Y = k | X = x)$.

(b) QDA

Expand the squared term in Eq. 2 with removing the assumption of constant class variance, we have

$$\log \pi_k - \log(\sqrt{2\pi}\sigma_k) - \frac{x^2}{2\sigma_k^2} + \frac{x\mu_k}{\sigma_k^2} - \frac{\mu_k^2}{2\sigma_k^2}. \quad (5)$$

Since the variances are different for different class, the final formula become:

$$-\frac{x^2}{2\sigma_k^2} + \frac{x\mu_k}{\sigma_k^2} - \frac{\mu_k^2}{2\sigma_k^2} + \log \pi_k - \log(\sigma_k), \quad (6)$$

which is a quadratic function in term of x .